THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 5–8: **Adolescence: A Stage of Growth and Change, Heat Transfer in Nature, Measurement of Time and Motion, Life Processes in Animals** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. An athlete runs 100 m in 12.5 s. The speed is: 1

(a) 0.125 m/s **(b)** 12.5 m/s **(c)** 8 m/s **(d)** 1250 m/s

2. If the diaphragm moves up, the volume of the chest cavity: 1

(a) stays same **(b)** increases **(c)** doubles **(d)** decreases

3. Adrenal glands lie above the: 1

(a) brain **(b)** kidneys **(c)** lungs **(d)** heart

4. Two layers of thin clothes keep you warmer than one thick layer because: 1

(a) thin clothes are heavy **(b)** air gets trapped between layers

(c) the clothes make their own heat **(d)** thick clothes are cold

5. Another line passes (0,0) and (2 s, 20 m). Compared with the first, this object is: 1

(a) slower **(b)** at rest **(c)** faster **(d)** equal

6. The vessels which have valves to prevent backflow of blood are: 1

(a) capillaries **(b)** all **(c)** arteries **(d)** veins

7. A child lacking insulin may develop: 1

(a) anaemia **(b)** scurvy **(c)** goitre **(d)** diabetes

8. When you place your hand beside a candle flame, you feel heat mainly due to: 1

(a) radiation **(b)** evaporation **(c)** conduction **(d)** convection

9. A speed of 25 m/s equals: 1

(a) 25 km/h **(b)** 90 km/h **(c)** 250 km/h **(d)** 9 km/h

10. Which pair is correct? 1

(a) amylase: proteins **(b)** lipase: starch

(c) pepsin: fats **(d)** pepsin: proteins

11. Which is not a symptom of puberty? 1

(a) pimples **(b)** change of voice

(c) loss of milk teeth at 5 **(d)** growth spurt

12. At night the sea stays warmer than land because: 1

(a) water cools slowly **(b)** land is warm

(c) sun shines **(d)** sea has no air

13. If a body doubles its speed, the time to cover the same distance becomes: 1

(a) half **(b)** double **(c)** quarter **(d)** same

14. Exhaled air has less ___ than inhaled air. 1

(a) oxygen **(b)** water vapour **(c)** carbon dioxide **(d)** heat

15. Pituitary gland lies at the base of the: 1

(a) brain **(b)** heart **(c)** stomach **(d)** kidney

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Why should a pendulum be released with a small angle? 2

17. Why do we pass less urine in summer? 2

18. How does the pituitary gland control puberty? 2

19. How does a hot-air balloon rise? 2

20. How long will a train at 90 km/h take to cover 225 km? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A train 200 m long passes a pole in 20 s. Find its speed in m/s and km/h. 3

22. Describe an experiment to show that exhaled air contains carbon dioxide. 3

23. Explain the difference between the endocrine and digestive glands. 3

24. A student heats a metal rod at the centre. Describe how heat spreads. 3

OR

Why is the speed of a vehicle in cities less than on highways? 3

25. Explain how a cockroach takes in air and delivers oxygen. 3

OR

Why should adolescents share worries with elders? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the importance of nutrition, sleep, hygiene and physical activity for adolescents with reasons. 5

OR

Explain how the combined effects of conduction, convection and radiation work in a thermos flask and what could happen if the vacuum is lost. 5

27. Draw and explain the distance-time graph of a car moving at 20 km/h for 2 h, resting for 1 h and then moving at 30 km/h for 2 h. 5

OR

Describe the structure and working of the human heart and the path of blood in the body. 5

END OF QUESTION PAPER — **Total: 50 Marks**